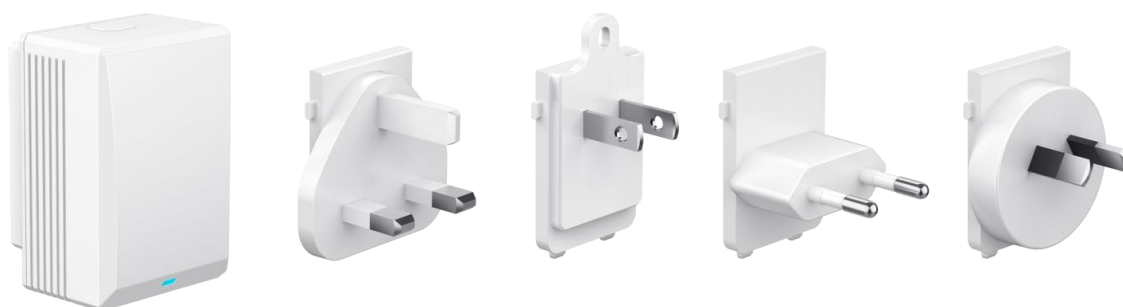




MOKOsmart Remote Gateway-32D

Communication Protocol

Version V1.0

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1. About This Manual

This manual describes the MOKO Remote Gateway-32D communication protocol, including BLE protocol used to configure the gateway and the MQTT protocol used to manage the gateway from you cloud.

This document can be applied in the following gateways, as there are some differences on the gateways hardware, their software and protocol have some minor differences, but most of the commands are the same.

Gateway Model	Main Features Description
MK107Pro-32D	➤ 2.4GHZ WIFI and BLE 5.0 ➤ WIFI security supports enterprise level (802.1X) ➤ Smart control for socket on/off
MKGW-mini 02-32D	➤ 2.4GHZ WIFI and BLE 5.0 ➤ WIFI security supports enterprise level (802.1X) ➤ No socket output

2. BLE Protocol

BLE function is enabled when device is in the configuration mode. In this mode, the BLE commands are executed from APP to the gateway, used to configure WIFI, MQTT and other settings for the gateway.

2.1 Bluetooth Advertising

In configuration mode, the gateway indicator flashes blue, and the Bluetooth advertises, can be discovered and connected by a BLE host. The advertising data format is as below:

Advertising packet:

Length	Data Type	Data
2	0X01	BLE Flags: 0X06 GeneralDiscoverable, BrEdrNotSupported
4	0X16	Service UUID (0XAA0B) and device type (0X60: MK107Pro-32D/ 0X70: MKGW-mini 02-32D)
16	0X09	Complete Local Name

Response packet:

Length	Data Type	Data
15	0XFF	Manufacturer data. Service UUID (0XAA0B) + BLE MAC (6 bytes)+ Firmware version(V x.x.x)

2.2 Service and Characteristic

Service(0XAA00)	
Characteristic	Function
0XAA00	Password verification (Write/notify)
0XAA01	Bluetooth disconnection type (Notify)
0XAA03	Configuration (Write/notify)

2.3 Command Format

BLE commands are divided into two types according to the data length. For one command, the max data length is 178 bytes. If the data length is greater than 178 bytes, it needs to be divided into several packets to send.

- Command format for data length is less than 178 bytes:

No.	Name	Description
1	Head	0XED
2	Flag	Properties flag. 0X00: read 0X01: write 0X02: notify
3	CMD id	Command id, 1 byte.
4	Length	Data length, 1 byte. Range: 0X01-0XB2
5	Data	Date value, max 178 bytes

Inquire: 0XED + 0X00 + CMD id+ 0X00

Reply: 0XED + 0X00+ CMD id+ length+ data

Setup: 0XED+ 0X01+ CMD id+ length+ data

Reply: 0XED+ 0X01+ CMD id+ 0X01+ 0X01/0X00 (0X01: success, 0X00: fail)

- Command format for data length is greater than 178 bytes:

No.	Name	Description
1	Head	0XEE
2	Flag	Properties flag. 0X00: read 0X01: write 0X02: notify
3	CMD id	Command id, 1 byte.
4	Packet_num	1 byte. To indicate the total packet number
5	Packet_seq	1 byte. To indicate packet sequence, start from 0
6	Length	Data length, 1 byte. range: 0X01-0XB0
7	Data	Date value, max 176 bytes

Inquire: 0XEE + 0X00 + CMD id +0X00

Reply: 0XEE + 0X00+ CMD id + packet_num + packet_seq + length+ data

Setup: 0XEE + 0X01+ CMD id + packet_num + packet_seq + length + data

Reply: 0XEE + 0X01+ CMD id + 0X01 + 0X01/0X00 (0X01: success, 0X00: fail)

Remark: The device will reply to each setup command, the middle replies tell user that the device receives the command, the last reply tells the setup result.

2.4 Password Verification (0XAA00)

It is under the characteristic 0XAA00.

After the gateway Bluetooth is connected, it requires to enter the password, the password is **Moko4321**.

APP send:

Head	Flag	CMD id	Data Length	Data Value
0XED	0X01	0X01	0X06-0X0A	6-10 bytes characters

Device respond:

Head	Flag	CMD id	Data Length	Data Value
0XED	0X01	0X01	0X01	0X01: success 0X00: Fail

APP send: ED 01 01 08 4D 6F 6B 6F 34 33 32 31 (password: Moko4321)

Device response: ED 01 01 01 01 (success)

2.5 BLE Disconnection Type (0XAA01)

It is under the characteristic 0XAA01. When Bluetooth disconnects, this message will be actively sent by the device.

Device send:

Head	Flag	CMD id	Data Length	Data Value
0XED	0X02	0X01	0X01	0X01: Disconnect due to password error or password verification time-out. 0X02: Disconnect due to communication time-out. (After Bluetooth is connected, if there is no communication for 10 minutes, Bluetooth will disconnect.)

Device send: ED 02 01 01 01 (Password verification time-out)

2.6 Configuration (0XAA03)

It is under the characteristic 0XAA03. The commands are sent from APP to the gateway to configure the gateway.

2.6.1 System parameters

2.6.1.1 Reboot device (0X01)

This command is to reboot the gateway.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X01	0X01	0X00	/

APP send: ED 01 01 00

Device respond: ED 01 01 01 01(success)

2.6.1.2 Connect network (0X02)

After MQTT and WIFI are configured, send this command, the gateway will connect to the WIFI and MQTT server.

Header	Flag	CMD ID	Data Length	Data Value
Write	0X01	0X02	0X00	/

APP send: ED 01 02 00
Device respond: ED 01 02 01 01(success)

2.6.1.3 BLE password (0X03)

To read and set the BLE connection password.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X03	0X06-0X0A	6-10 characters

Inquire:

APP send: ED 00 03 00
Device respond: ED 00 03 08 31 32 33 34 35 36 37 38 (Password:12345678)

Setup:

APP send: ED 01 03 08 31 32 33 34 35 36 37 38 (Password:12345678)
Device respond: ED 01 03 01 01 (success)

2.6.1.4 Reset device by button (0X04)

To read and set the way to reset gateway. By default, pressing button for 10 seconds in 1 minute after powered will reset the gateway, it can be changed to pressing button any time.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0x04	0X01	0: Press button in 1 minute after powered 1: Press button any time

Inquire:

APP send: ED 00 04 00
Device respond: ED 01 04 01 01 (01: press button in 1 minutes after powered)

Setup:

APP send: ED 01 04 01 02 (Press ant time)
Device respond: ED 01 04 01 01 (success)

2.6.1.5 Device name (0X05)

To read the gateway advertising name.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00	0x05	0X01-0X14	1-20 characters

Inquire:

APP send: ED 00 05 00

Device respond: ED 00 05 0A 4D 4B 31 31 30 2D 35 36 32 36 (Device name: MK110-5626)

2.6.1.6 Product model (0X06)

To read the product model.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00	0x06	0X01-0X14	1-20 characters

Inquire:

APP send: ED 00 06 00

Device respond: ED 00 06 05 4D 4B 31 31 30 (Product model: MK110)

2.6.1.7 Hardware version (0X07)

To read the hardware version.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00	0x07	0X01-0X14	1-20 characters

Inquire:

APP send: ED 00 07 00

Device respond: ED 00 07 06 56 31 2E 30 2E 33 (Hardware version: V1.0.3)

2.6.1.8 Manufacturer (0X08)

To read the manufacturer information.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00	0x08	0X01-0X14	1-20 characters

Inquire:

APP send: ED 00 08 00

Device respond: ED 00 08 14 4D 4F 4B 4F 20 54 45 43 48 4E 4F 4C 4F 47 59 20 4C 54 44 2E (Manufacturer: MOKO TECHNOLOGY LTD.)

2.6.1.9 BLE MAC (0X09)

To read the BLE MAC address.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00	0x09	0X06	6 bytes Mac address

Inquire:

APP send: ED 00 09 00

Device respond: ED 00 09 06 4C 11 AE 8B E6 26 (BLE MAC: 4C 11 AE 8B E6 26)

2.6.2.10 WIFI MAC (0X0A)

To read the WIFI STA MAC address.

Header	Flag	CMD ID	Data Length	Data Value
OXED	0X00	0x0A	0X06	6 bytes Mac address

Inquire:

APP send: ED 00 0A 00

Device respond: ED 00 0A 06 4C 11 AE 8B E6 24 (WIFI MAC: 4C 11 AE 8B E6 24)

2.6.1.11 NTP Switch (0X10)

To read and set the NTP switch.

Header	Flag	CMD ID	Data Length	Data Value
OXED	0X00/0X01	0X10	0X01	0/1 0: on, 1: off

Inquire:

APP send: ED 00 10 00

Device respond: ED 00 10 01 01 (NTP switch on)

Setup:

APP send: ED 01 10 01 00 (NTP switch off)

Device respond: ED 01 10 01 01 (success)

2.6.1.12 NTP Server (0X11)

To read and set the NTP server for the gateway.

The gateway has three default NTP servers, it allows users to setup one NTP server. If users do not set the NTP server, the gateway will use the three default servers.

Header	Flag	CMD ID	Data Length	Data Value
OXED	0X00/0X01	0X11	0X00-0X40	0-64 characters

Inquire:

APP send: ED 00 11 00

Device respond: ED 00 11 0D 77 77 77 2E 62 61 69 64 75 2E 63 6F 6D (NTP server: www.baidu.com)

Setup:

APP send: ED 01 11 0D 77 77 77 2E 62 61 69 64 75 2E 63 6F 6D (NTP server:

www.baidu.com)

Device respond: ED 01 11 01 01 (success)

2.6.1.13 Timezone(0X12)

To read and set the timezone for the gateway.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X12	0X01	Signed number. -24~28, step by half timezone,

Inquire:

APP send: ED 00 12 00

Device respond: ED 00 12 01 10 (Timezone: UTC+8)

Setup:

APP send: ED 01 12 01 10

Device respond: ED 01 12 01 E8 (Timezone: UTC-12)

2.6.1.14 Firmware version(0X13)

To read the gateway firmware version.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00	0X13	0X01	1-20 characters

Inquire:

APP send: ED 00 13 00

Device respond: ED 00 13 06 56 31 2E 30 2E 31 (Firmware version: V1.0.3)

2.6.1.15 ON/OFF control by button (0X15)

Only works on MK107Pro-32D gateway.

MK107Pro-32D gateway has an AC socket, the on/off status can be controlled by pressing button. This command is to enable/disable the feature to changes AC output status by pressing button.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X01	0X15	0X01	0X00: disable 0X01: enable

Setup:

APP send: ED 01 15 01 01 (enable)

Device respond: ED 01 15 01 01(success)

2.6.2 MQTT parameters

2.6.2.1 MQTT Host (0X20)

To read and set the MQTT host.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X20	0X01-0X40	1-64 characters. Default is 47.104.81.55

Inquire:

APP send: ED 00 20 00

Device respond: ED 00 20 0C 34 37 2E 31 30 34 2E 38 31 2E 35 35 (Host: 47.104.81.55)

Setup:

APP send: ED 01 20 0C 34 37 2E 31 30 34 2E 38 31 2E 35 35 (Host: 47.104.81.55)

Device respond: ED 01 20 01 01 (success)

2.6.2.2 MQTT Port (0X21)

To read and set the MQTT port.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X21	0X02	1-65535 Default is 1883

Inquire:

APP send: ED 00 21 00

Device respond: ED 00 20 02 07 5B (Port: 1883)

Setup:

APP send: ED 00 21 02 07 5B (Port: 1883)

Device respond: ED 01 21 01 01 (success)

2.6.2.3 Client id (0X22)

To read and set the client id.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X22	1-64	Characters. Default is wifi mac address

Inquire:

APP send: ED 00 22 00
Device respond: ED 00 22 04 4D 4B 47 57 (Client id: MKGW)

Setup:

APP send: ED 01 22 04 4D 4B 47 57 (Client id: MKGW)
Device respond: ED 01 22 01 01 (success)

2.6.2.4 MQTT Username(0X23)

To read and set the MQTT username. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
0XEE	0X00/ 0X01	0X23	1 byte	1 byte	0X00-0XB0	0-176 bytes

Inquire:

APP send: EE 00 23 00
Device respond: EE 00 23 01 00 04 30 31 32 33 (username: 0123)

Setup:

APP send: EE 01 23 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)
APP send: EE 01 23 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)
Device respond: ED 01 23 01 01 (success)

2.6.2.5 MQTT Password(0X24)

To read and set the MQTT password. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
0XEE	0X00/ 0X01	0X24	1 byte	1 byte	0X00-0XB0	0-176 bytes

Inquire:

APP send: EE 00 24 00
Device respond: EE 00 24 01 00 04 30 31 32 33 (password: 0123)

Setup:

APP send: EE 01 24 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)
APP send: EE 01 24 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)
Device respond: ED 01 24 01 01 (success)

2.6.2.6 Clean session(0X25)

To read and set the MQTT clean session.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X25	0X01	0/1

Inquire:

APP send: ED 00 25 00

Device respond: ED 00 25 01 01 (Yes)

Setup:

APP send: ED 01 25 01 01 (Yes)

Device respond: ED 01 25 01 01 (success)

2.6.2.7 Keep alive(0X26)

To read and set the MQTT clean session.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X26	0X01	10-120, unit: second Default is 60

Inquire:

APP send: ED 00 26 00

Device respond: ED 00 26 01 3C (Keep alive: 60)

Set command:

APP send: ED 01 26 01 3C (Keep alive: 60)

Device respond: ED 01 26 01 01 (success)

2.6.2.8 Qos (0X27)

To read and set the MQTT Qos.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X27	0X01	0/1/2

Inquire:

APP send: ED 00 27 00

Device respond: ED 00 27 01 00 (Qos: 0)

Setup:

APP send: ED 01 27 01 01 (Qos: 1)

Device respond: ED 01 27 01 01 (success)

2.6.2.9 Subscribe topic (0X28)

To read and set the gateway subscribe topic.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X28	0X01-0X80	1-128 characters. Default is /MK110 Plus 03/mac/receive

Inquire:

APP send: ED 00 28 00

Device respond: ED 00 28 04 31 32 33 34 (subscribe topic: 1234)

Setup:

APP send: ED 01 28 04 31 32 33 34 (subscribe topic: 1234)

Device respond: ED 01 28 01 01 (success)

2.6.2.10 Publish topic (0X29)

To read and set the gateway subscribe topic.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X29	0X01-0X80	1-128 characters. Default is /MK110 Plus 03/mac/send

Inquire:

APP send: ED 00 29 00

Device respond: ED 00 29 04 34 33 32 31 (publish topic: 4321)

Setup:

APP send: ED 01 29 04 34 33 32 31 (publish topic: 4321)

Device respond: ED 01 29 01 01 (success)

2.6.2.11 LWT switch (0X2A)

To read and set the MQTT LWT switch.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X2A	0X01	0/1

Inquire:

APP send: ED 00 2A 00

Device respond: ED 00 2A 01 00 (LWT switch: off)

Setup:

APP send: ED 01 2A 01 01 (LWT switch: on)
Device respond: ED 01 2A 01 01 (success)

2.6.2.12 LWT Qos (0X2B)

To read and set the MQTT LWT Qos.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X2B	0X01	0/1/2

Inquire:

APP send: ED 00 2B 00
Device respond: ED 00 2B 01 00 (Qos: 0)

Setup:

APP send: ED 01 2B 01 01 (Qos: 1)
Device respond: ED 01 2A 01 01 (success)

2.6.2.13 LWT Retain (0X2C)

To read and set the MQTT LWT retain.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X2C	0X01	0/1

Inquire:

APP send: ED 00 2C 00
Device respond: ED 00 2C 01 00 (retain: 1)

Setup:

APP send: ED 01 2C 01 01 (retain: 1)
Device respond: ED 01 2C 01 01 (success)

2.6.2.14 LWT Topic (0X2D)

To read and set the MQTT LWT Topic.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X2D	0X01-0X80	1-128 characters. Default is /MK110 Plus 03/mac/receive

Inquire:

APP send: ED 00 2D 00
Device respond: ED 00 2D 18 2F 4D 4B 31 31 30 2F 34 63 31 31 61 65 38 62 65 36

32 34 2F 73 65 6E 64 (LWT topic: /MK110/4c11ae8be624/send)

Setup:

APP send: ED 01 2D 18 2F 4D 4B 31 31 30 2F 34 63 31 31 61 65 38 62 65 36 32 34 2F 73 65 6E 64 (LWT topic: /MK110/4c11ae8be624/send)

Device respond: ED 01 2D 01 01 (success)

2.6.2.15 LWT Payload (0X2E)

To read and set the MQTT LWT payload.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X2E	0X01-0X80	1-128 characters. Default is {"msg_id" : 3999, "device_info" : {"mac":"XXXXXXXXXXXXX"}, "data" : {}}

Inquire:

APP send: ED 00 2E 00

Device respond: ED 00 2E 07 4F 46 46 4C 49 4E 45 (LWT payload: offline)

Setup:

APP send: ED 01 2E 07 4F 46 46 4C 49 4E 45 (LWT payload: offline)

Device respond: ED 01 2E 01 01 (success)

2.6.2.16 MQTT Encryption (0X2F)

To read and set the MQTT encryption.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X2F	0X01	0-3 0: No encryption 1: CA signed server certificate 2: CA certificate 3: Self signed certificates

Inquire:

APP send: ED 00 2F 00

Device respond: ED 00 2F 01 01 (no encryption)

Setup:

APP send: ED 01 2F 01 01 (CA signed server certificate)

Device respond: ED 01 2F 01 01 (success)

2.6.2.17 MQTT CA certificate (0X30)

To send CA certificate to the gateway. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
OXEE	OX01	OX30	1 byte	1 byte	OX00-OXB0	0-176 bytes

APP send: EE 01 30 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)

APP send: EE 01 30 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)

Device respond: ED 01 30 01 01 (success)

2.6.2.18 MQTT client certificate (0X31)

To send client certificate to the gateway. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
OXEE	OX01	OX31	1 byte	1 byte	OX00-OXB0	0-176 bytes

APP send: EE 01 31 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)

APP send: EE 01 31 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)

Device respond: ED 01 31 01 01 (success)

2.6.2.19 MQTT client key (0X32)

To send client key to the gateway. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
OXEE	OX01	OX32	1 byte	1 byte	OX00-OXB0	0-176 bytes

APP send: EE 01 32 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)

APP send: EE 01 32 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)

Device respond: ED 01 32 01 01 (success)

2.6.3 WiFi parameters

2.6.3.1 WIFI security (0X40)

To read and set the WIFI encryption.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X40	0X01	0-1 0: personal 1: enterprise

Inquire:

APP send: ED 00 40 00

Device respond: ED 00 40 01 01 (WIFI security: enterprise)

Setup:

APP send: ED 01 40 01 01 (WIFI security: enterprise)

Device respond: ED 01 40 01 01 (success)

2.6.3.2 WIFI SSID (0X41)

To read and set the WIFI SSID.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X41	0X01-0X20	1-32 characters

Inquire:

APP send: ED 00 41 00

Device respond: ED 00 41 04 31 32 33 34 (WIFI ssid: 1234)

Setup:

APP send: ED 01 41 04 31 32 33 34 (WIFI ssid: 1234)

Device respond: ED 01 41 01 01 (success)

2.6.3.3 WIFI Password (0X42)

To read and set the WIFI password.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X42	0X00-0X40	0-64 characters

Inquire:

APP send: ED 00 42 00

Device respond: ED 00 42 08 31 32 33 34 35 36 37 38 (WIFI password: 12345678)

Setup:

APP send: ED 01 42 08 31 32 33 34 35 36 37 38 (WIFI password: 12345678)
Device respond: ED 01 42 01 01 (success)

2.6.3.4 WIFI EAP type (0X43)

To read and set the WIFI EAP type for the enterprise WiFi.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X43	0X01	0: PEAP-MSCHAPV2 1: TTLS-MSCHAPV2 2: TLS

Inquire:

APP send: ED 00 43 00
Device respond: ED 00 43 01 00 (EAP type: PEAP-MSCHAPV2)

Setup:

APP send: ED 01 43 01 00 (EAP type: PEAP-MSCHAPV2)
Device respond: ED 01 43 01 01 (success)

2.6.3.5 WIFI EAP username(0X44)

To read and set the EAP username.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X44	0X00-0X20	0-32 characters. Valid when EAP type is 0 or 1

Inquire:

APP send: ED 00 44 00
Device respond: ED 00 44 04 31 32 33 34(EAP username: 1234)

Setup:

APP send: ED 01 44 04 31 32 33 34(EAP username: 1234)
Device respond: ED 01 44 01 01 (success)

2.6.3.6 WIFI EAP password (0X45)

To read and set the EAP password.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X45	0X00-X040	0-64 characters. Valid when EAP type is 0 or 1

Inquire:

APP send: ED 00 45 00

Device respond: ED 00 45 04 31 32 33 34(EAP password: 1234)

Setup:

APP send: ED 01 45 04 31 32 33 34(EAP password: 1234)

Device respond: ED 01 45 01 01 (success)

2.6.3.7 WIFI EAP Domain ID (0X46)

To read and set the EAP domain id.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X46	0X00-0X40	0-64 characters. Valid when EAP type is 2

Inquire:

APP send: ED 00 46 00

Device respond: ED 00 46 04 31 32 33 34(Domain id: 1234)

Setup:

APP send: ED 01 46 04 31 32 33 34(Domain id: 1234)

Device respond: ED 01 46 01 01 (success)

2.6.3.8 WIFI EAP verify server (0X47)

To read and set EAP verify server option.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X47	0X01	0/1

Inquire:

APP send: ED 00 47 00

Device respond: ED 00 47 01 01(Verify server)

Setup:

APP send: ED 01 47 01 01 (Verify server)

Device respond: ED 01 47 01 01 (success)

2.6.3.9 WIFI CA certificate(0X48)

To send WIFI CA certificate to the gateway. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
OXEE	OX01	OX48	1 byte	1 byte	OX00-OXB0	0-176 bytes

APP send: EE 01 48 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)

APP send: EE 01 48 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)

Device respond: ED 01 48 01 01 (success)

2.6.3.10 WIFI client certificate (OX49)

To send WIFI client certificate to the gateway. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
OXEE	OX01	OX49	1 byte	1 byte	OX00-OXB0	0-176 bytes

APP send: EE 01 49 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)

APP send: EE 01 49 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)

Device respond: ED 01 49 01 01 (success)

2.6.3.11 WIFI client key (OX4A)

To send WIFI client key to the gateway. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
OXEE	OX01	OX4A	1 byte	1 byte	OX00-OXB0	0-176 bytes

APP send: EE 01 4A 02 00 B0 30 31 32 33 34 35 36..... (Total 2 packets, the first packet)

APP send: EE 01 4A 02 01 B0 30 31 32 33 34 35 36..... (Total 2 packets, the second packet)

Device respond: ED 01 4A 01 01 (success)

2.6.3.12 DHCP switch (OX4B)

To read and set the DHCP switch.

Header	Flag	CMD ID	Data Length	Data Value
OXED	OX00/OX01	OX4B	OX01	0/1

				0: statics ip 1: DHCP
--	--	--	--	--------------------------

Inquire:

APP send: ED 00 4B 00

Device respond: ED 00 4B 01 01 (DHCP switch: on)

Setup:

APP send: ED 01 4B 01 01 (DHCP switch: on)

Device respond: ED 01 4B 01 01 (success)

2.6.3.13 IP (0X4C)

To read and set the ip when using statics ip.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X4C	0X10	1-4 byte: ip address 5-8 byte: mask 9-12 byte: gateway 13-16 byte: DNS

Inquire:

APP send: ED 00 4C 00

Device respond: ED 00 4C 10 00 00 00 00 00 00 00 00 00 00 00 00 00 (ip: 0.0.0.0, mask: 0.0.0.0, gateway: 0.0.0.0, DNS: 0.0.0.0)

Setup:

APP send: ED 01 4C 10 C0 A8 00 02 FF FF FF 00 C0 A8 00 01 C0 A8 00 01 (ip: 192.168.0.2, mask: 255.255.255.0, gateway: 192.168.0.1, DNS: 192.168.0.1)

Device respond: ED 01 4C 01 01 (success)

2.6.4 BLE Scan parameters

2.6.4.1 RSSI filter (0X60)

To read and set the RSSI filter.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X60	0X01	-127~0, dBm

Inquire:

APP send: ED 00 60 00

Device respond: ED 00 60 01 C4 (rssi: -60 dBm)

Setup:

APP send: ED 01 60 01 C4 (rssi: -60 dBm)

Device respond: ED 01 60 01 01 (success)

2.6.4.2 Mac filter (0X64)

To read and set the mac address filter.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X64	0X00-0X46	Length+MAC, the length is 1 byte, the mac is 1-6 byte. Can set at most 10 groups mac address.

Inquire:

APP send: ED 00 64 00

Device respond: ED 00 64 07 02 AA B4 03 B5 F4 32 (the first mac is "AAB4", the second mac is "B5F432")

Setup:

APP send: ED 01 64 07 02 AA B4 03 B5 F4 32 (the first mac is "AAB4", the second mac is "B5F432")

Device respond: ED 01 64 01 01 (success)

2.6.4.3 ADV name filter (0X67)

To read and set the advertising name filter. This command needs to use the sub-packet format.

Header	Flag	CMD ID	packet_num	packet_seq	length	Value
0XEE	0X00/0X01	0X67	1 byte	1 byte	0X00-0XB0	Length+name, the length is 1 byte, the name is 1-20 byte. Can set at most 10 groups adv names

Inquire:

APP send: EE 00 67 00

Device respond: EE 00 67 01 00 08 04 4D 4F 4B 4F 02 30 30 (the first name is "MOKO", the second name is "00")

Setup:

APP send: EE 01 67 01 00 08 04 4D 4F 4B 4F 02 30 30 (the first name is "MOKO",

the second name is "00")

Device respond: EE 01 67 01 01 (success)

2.6.5 Advertise iBeacon parameters

The gateway supports iBeacon advertising. This command is used to inquire and setup the advertising parameters.

2.6.5.1 Advertising switch (0X70)

To turn on/off advertising iBeacon.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X70	0X01	0: off 1: on

Inquire:

APP send: ED 00 70 00

Device respond: ED 00 70 01 01(advertising switch on)

Setup:

APP send: ED 01 70 01 01 (turn on advertising)

Device respond: ED 01 70 01 01 (success)

2.6.5.2 iBeacon major (0X71)

To inquire and setup the advertising iBeacon major.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X71	0X02	Range: 0-65535

Inquire:

APP send: ED 00 71 00

Device respond: ED 00 71 02 00 01(major: 1)

Setup:

APP send: ED 01 71 02 00 01 (major: 1)

Device respond: ED 01 71 01 01 (success)

2.6.5.3 iBeacon minor (0X72)

To inquire and setup the advertising iBeacon minor.

Header	Flag	CMD ID	Data Length	Data Value
--------	------	--------	-------------	------------

OXED	OX00/OX01	OX72	OX02	Range: 0-65535
------	-----------	------	------	----------------

Inquire:

APP send: ED 00 72 00

Device respond: ED 00 72 02 00 00(minor: 0)

Setup:

APP send: ED 01 72 02 00 01 (major:1)

Device respond: ED 01 72 01 01 (success)

2.6.5.4 iBeacon UUID (OX73)

To inquire and setup the advertising iBeacon UUID.

Header	Flag	CMD ID	Data Length	Data Value
OXED	OX00/OX01	OX73	OX10	16 bytes

Inquire:

APP send: ED 00 73 00

Device respond: ED 00 73 10 00 11 22 33 44 55 66 77 88 99 AA BB CC DD EE FF
(UUID: 112233445566778899AABBCCDDEEFF)

Setup:

APP send: ED 01 73 10 00 11 22 33 44 55 66 77 88 99 AA BB CC DD EE FF (UUID: 112233445566778899AABBCCDDEEFF)

Device respond: ED 01 73 01 01 (success)

2.6.5.5 Advertising interval (OX74)

To inquire and setup the advertising interval.

Header	Flag	CMD ID	Data Length	Data Value
OXED	OX00/OX01	OX74	OX01	1-100, unit: 100ms Default: 1

Inquire:

APP send: ED 00 74 00

Device respond: ED 00 74 01 01(interval: 100ms)

Setup:

APP send: ED 01 74 01 05 (interval: 500ms)

Device respond: ED 01 74 01 01 (success)

2.6.5.6 Tx power (0X75)

To inquire and setup the advertising TX power.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X75	0X01	Range: 0-15 0: -24 dBm 1: -21 dBm 2: -18 dBm 3: -15 dBm 4: -12 dBm 5: -9 dBm 6: -6 dBm 7: -3 dBm 8: 0 dBm 9: 3 dBm 10: 6 dBm 11: 9 dBm 12: 12 dBm 13: 15 dBm 14: 18 dBm 15: 21 dBm default: 9

Inquire:

APP send: ED 00 75 00

Device respond: ED 00 75 01 09 (Tx power: 3 dBm)

Setup:

APP send: ED 01 75 01 0F (Tx power: 21 dBm)

Device respond: ED 01 75 01 01 (success)

2.6.5.7 RSSI@1m (0X76)

To inquire and setup the iBeacon RSSI@1m.

Header	Flag	CMD ID	Data Length	Data Value
0XED	0X00/0X01	0X76	0X01	Range: -100~0, dbm Default: -59

Inquire:

APP send: ED 00 76 00

Device respond: ED 00 76 01 C5 (RSSI@1m: -59)

Setup:

APP send: ED 01 76 01 C5 (RSSI@1m: -59)
Device respond: ED 01 76 01 01 (success)

3. MQTT Protocol

After the gateway is connected with MQTT broker, all communications are based on MQTT commands.

3.1 Commands Format

The commands are sent from cloud to configure or inquire the gateway parameters, the gateway will reply to each command.

The command is using message id to distinguish different actions. The message id starts from 1000 in the setup commands, starts from 2000 in the inquire commands, starts from 3000 in the notification commands.

3.1.1 Setup gateway parameters

Send from cloud

```
{
  "msg_id": 1XXX,      //Message id, to distinguish different commands. The
                        //message starts from 1000 in setup command
  "device_info":{
    "mac":"112233445566"    //The MAC address of the gateway
  },
  "data":{
    "XX" : XX              //Parameter name and value
  }
}
```

Gateway reply

```
{
  "msg_id":1XXX,      //Message id
  "device_info":{
    "mac":"112233445566"    //The MAC address of the gateway
  },
  "result_code":X,      // Return code
  "result_msg":"XX"      // Description of the code
}
```

3.1.2 Inquire gateway parameters

Send from cloud

```
{
  "msg_id": 2XXX,    //Message id, the message starts from 2000 in inquire
                    //command
  "device_info":{
    "mac":"112233445566"    //The MAC address of the gateway
  }
}
```

Gateway reply

```
{
  "msg_id":2XXX,    //Message id
  "device_info":{
    "mac":"112233445566"    //The MAC address of the gateway
  },
  "data":{
    "XX":XX,    // Parameter name and value
  }
}
```

3.1.3 Gateway notify data to cloud

Gateway will actively report data to cloud in some situations. For example, reports BLE scanning data, reports OTA result and so on.

The reporting data format is as below:

```
{
  "msg_id":3XXX,    //Message id. The message id starts from 3000 in
                    //notification command
  "device_info":{
    "mac":"112233445566"    //The MAC address of the gateway
  },
  "data":{
    "XX" : XX    //Data and value
  }
}
```

3.2 Result Code Description

When a command is sent to the gateway, the gateway will reply to each command.

There is a return code included in the reply to tell users the setup result. Following are the result code and their descriptions:

result_code	result_msg	description
0	success	Receive and parse the message successfully
1	length error	The data length is out of range
2	type error	The data type doesn't conform the rule
3	range error	The data value is out of range
4	no object error	The parameter name is not found

3.3 Command List

In the setup command, the message id is 1XXX, in the inquire command, the message id is 2XXX, in the notify command, the message is 3XXX, we define the XXX is the message id offset.

3.3.1 System parameters

3.3.1.1 Reboot device (000)

This command is used to reboot the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	0	reset	number. 0: reboot

Setup:

```
{
  "msg_id": 1000,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "reset": 0
  }
}
```

Device reply:

```
{
  "msg_id": 1000,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

```
}
```

3.3.1.2 Reset gateway by button (001)

This command is used to define the button operation triggers to reset the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	1	key_reset_type	number. 0-1 0: Press button in 1 minute after powered 1: Press button any time

Inquire:

```
{
  "msg_id": 2001,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2001,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "key_reset_type": 1
  }
}
```

Setup:

```
{
  "msg_id": 1001,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "key_reset_type": 1
  }
}
```

Device reply:

```
{
  "msg_id": 1001,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

```

    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.1.3 Device information (002)

To read device information of the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Read	2	ble_mac	string
		wifi_mac	string
		device_name	string
		product_model	string
		company_name	string
		hardware_version	string
		software_version	string
		firmware_version	string

Inquire:

```

{
  "msg_id": 2002,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}

```

Device reply:

```

{
  "msg_id": 2002,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "ble_mac": "4c11ae8be626",
    "wifi_mac": "4c11ae8be624",
    "device_name": "MK107Pro-E626",
    "product_model": "MK107Pro",
    "company_name": "333",
    "hardware_version": "V0.1",
    "software_version": "V4.4.3",
    "firmware_version": "V1.0.1"
  }
}

```

```
}  
}
```

3.3.1.4 Network report interval (003)

This command is used to define the time interval of gateway reporting network status to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	3	report_interval	number. Range: 0 or 30-86400, second Default: 30

Inquire:

```
{  
  "msg_id": 2003,  
  "device_info": {  
    "mac": "4c11ae8be624"  
  }  
}
```

Device reply

```
{  
  "msg_id": 2003,  
  "device_info": {  
    "mac": "4c11ae8be624"  
  },  
  "data": {  
    "report_interval": 30  
  }  
}
```

Setup:

```
{  
  "msg_id": 1003,  
  "device_info": {  
    "mac": "4c11ae8be624"  
  },  
  "data": {  
    "report_interval": 30  
  }  
}
```

Device reply:

```
{  
  "msg_id": 1003,
```

```

    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.1.5 Notify network status (004)

The gateway will automatically report the network status after it connects to server.

Flag	Msg id offset	Parameter name	Parameter value
Notify	4	timestamp	number. Unix timestamp
		timezone	number. -24~28, step by half timezone.
		wifi_rssi	number.

Device notify:

```

{
    "msg_id": 3004,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timestamp": 1689219704,
        "timezone": 0,
        "wifi_rssi":
    }
}

```

3.3.1.6 Reconnect timeout (005)

The gateway will automatically reboot once if it fails to reconnect to server in the configured timeout. This command is used to configure the timeout.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	5	timeout	number. Range: 0-1440, minute. Default: 3

Inquire:

```

{
    "msg_id": 2005,
    "device_info": {

```

```

        "mac": "4c11ae8be624"
    }
}

```

Device reply:

```

{
    "msg_id": 2005,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timeout": 3
    }
}

```

Setup:

```

{
    "msg_id": 1005,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "timeout": 3
    }
}

```

Device reply

```

{
    "msg_id": 1005,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.1.7 WIFI Firmware upgrade (006)

This command is to send OTA information to the gateway, the gateway will download firmware upgrade file from the HTTP server for upgrading.

Flag	Msg id offset	Parameter name	Parameter value
Write	6	firmware_url	string. 1-256 characters

Setup:

```

{

```

```

"msg_id": 1006,
"device_info": {
  "mac": "4c11ae8be624"
},
"data": {
  "firmware_url": "http://47.104.172.255:8080/esp32.bin"
}

```

Device reply:

```

{
  "msg_id": 1006,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
}

```

3.3.1.8 WIFI firmware upgrade result (007)

The gateway will actively report the upgrade result after OTA is completed.

Flag	Msg id offset	Parameter name	Parameter value
Notify	7	result_code	number. -1~1 -1: under OTA 0: failed 1: success

Device notify:

```

{
  "msg_id": 3007,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "result_code": 1
  }
}

```

3.3.1.9 NTP setting (008)

To inquire and setup the NTP function switch and NTP server for the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	8	switch_value	number. 0/1 default: 1

		server	string 0-64 characters
--	--	--------	---------------------------

Inquire:

```
{
  "msg_id": 2008,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2008,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "server": "www.baidu.com"
  }
}
```

Setup:

```
{
  "msg_id": 1008,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "server": "www.baidu.com"
  }
}
```

Device reply:

```
{
  "msg_id": 1008,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.1.10 System time (009)

To inquire and setup the UTC time and timezone for the gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	9	timestamp	number. Unix timestamp
		timezone	number. -24~28, step by half timezone

Inquire:

```
{
  "msg_id": 2009,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2009,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "timestamp": 1681,
    "timezone": 0
  }
}
```

Setup:

```
{
  "msg_id": 1009,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "timestamp": 1681,
    "timezone": 0
  }
}
```

Device reply:

```
{
  "msg_id": 1009,
  "device_info": {
```

```

    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}

```

3.3.1.11 Communicate timeout (010)

After the gateway connects to a Beacon, if there is no communication in the timeout time, the gateway will automatically disconnect from the Beacon. This command is to inquire and setup the communication timeout.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	10	timeout	number. Range: 0-30, minute. Default: 10

Inquire:

```

{
  "msg_id": 2010,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}

```

Device reply:

```

{
  "msg_id": 2010,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "timeout": 10
  }
}

```

Setup:

```

{
  "msg_id": 1010,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "timeout": 10
  }
}

```

Device reply:

```
{
  "msg_id": 1010,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.1.12 LED setting (011)

To inquire and setup the LED switch under different device status.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	11	server_connecting_led	number. 0-1 0: off, 1: on
		server_connected_led	number. 0-1 0: off, 1: on

Inquire:

```
{
  "msg_id": 2011,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2011,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "server_connecting_led": 1,
    "server_connected_led": 1
  }
}
```

Setup:

```
{
  "msg_id": 1011,
  "device_info": {
    "mac": "4c11ae8be624"
  },
}
```

```

    "data": {
      "server_connecting_led": 1,
      "server_connected_led": 1
    }
  }
}

```

Device reply:

```

{
  "msg_id": 1011,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}

```

3.3.1.13 OTA or certificate download status (012)

To inquire OTA status or certificate download status.

Flag	Msg id offset	Parameter name	Parameter value
Read	12	status	number.0-1 0: not under OTA or updating 1: under OTA or downloading

Inquire:

```

{
  "msg_id": 2012,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}

```

Device reply:

```

{
  "msg_id": 2012,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "status": 0
  }
}

```

3.3.1.14 Reset device (013)

To reset the gateway. After reset, the gateway parameters will be restored to factory

settings.

Flag	Msg id offset	Parameter name	Parameter value
Write	13	factory_reset	number. 0: reset

Setup:

```
{
  "msg_id": 1013,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "factory_reset": 0
  }
}
```

Device reply:

```
{
  "msg_id": 1013,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.1.15 Notify reset operation (014)

When the gateway is reset by pressing button, the gateway will actively report a notification.

Flag	Msg id offset	Parameter name	Parameter value
Notify	14	/	/

Device notify:

```
{
  "msg_id": 3014,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
  }
}
```

3.3.1.16 Output switch (015)

Only works on MK107Pro-32D gateway.

The gateway has the AC output, this command is to inquire and setup the AC output status.

Flag	Msg id offset	Parameter name	Parameter value
Read/write	15	switch_value	number. 0-1 0: off, 1: on

Inquire:

```
{
  "msg_id": 2015,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2015,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
  }
}
```

Setup:

```
{
  "msg_id": 1015,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
  }
}
```

Device reply:

```
{
  "msg_id": 1015,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
}
```

```
"result_msg": "success"
}
```

3.3.1.17 Switch output by button (016)

Only works on MK107Pro-32D gateway.

The gateway has a physical button which can be used to switch the AC output. By default, switching AC output by button is disabled, it can be set to enabled.

This command is used to inquire and setup the function switch.

Flag	Msg id offset	Parameter name	Parameter value
Read/write	16	switch_value	number. 0-1 0: disable, 1: enable

Inquire:

```
{
  "msg_id": 2016,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2016,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 0,
  }
}
```

Setup:

```
{
  "msg_id": 1016,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
  }
}
```

Device reply:

```
{
```

```

    "msg_id": 1016,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.1.18 BLE Firmware upgrade (017)

This command is to send OTA information to the gateway, the gateway will download firmware upgrade file from the HTTP server for upgrading.

Flag	Msg id offset	Parameter name	Parameter value
Write	17	firmware_url	string. 1-256 characters

Setup:

```

{
    "msg_id": 1017,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "firmware_url": "http://47.104.172.255:8080/esp32_c3.bin"
    }
}

```

Device reply:

```

{
    "msg_id": 1017,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
}

```

3.3.1.19 BLE firmware upgrade result (018)

The gateway will actively report the upgrade result after OTA is completed.

Flag	Msg id offset	Parameter name	Parameter value
Notify	18	result_code	number. -1~1 -1: under OTA 0: failed 1: success

Device notify:

```
{
  "msg_id": 3018,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "result_code": 1
  }
}
```

3.3.2 WIFI settings

To read and change the WIFI settings for the gateway. After sending these commands, and send the reboot command (msg id: 1000), the gateway will reboot and reconnect using the new WIFI settings.

3.3.2.1 WIFI parameters (020)

Flag	Msg id offset	Parameter name	Parameter value
Write/read	20	security_type	number. 0-1 0: personal 1: enterprise
		ssid	string 1-32 characters
		passwd	string 0-64 characters
		eap_type	number. 0-2 0: PEAP-MSCHAPV2 1: TTLS-MSCHAPV2 2: TLS
		eap_id	string 0-64 characters
		eap_username	string 0-32 characters
		eap_passwd	string 0-64 characters
		eap_verify_server	number. 0-1 Valid when eap type is 0 Or 1

Inquire:

```
{
  "msg_id": 2020,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2020,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "security_type": 0,
    "ssid": "moko1234",
    "passwd": "12345678",
    "eap_type": 0,
    "eap_id": "",
    "eap_username": "",
    "eap_passwd": "",
    "eap_verify_server": 0
  }
}
```

Setup:

```
{
  "msg_id": 1020,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "security_type": 0,
    "ssid": "moko1234",
    "passwd": "12345678",
    "eap_type": 0,
    "eap_id": "",
    "eap_username": "",
    "eap_passwd": "",
    "eap_verify_server": 0
  }
}
```

Device reply:

```
{
  "msg_id": 1020,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.2.2 EAP certificates (021)

To change the EAP certificates, valid when the WIFI is in enterprise security.

Flag	Msg id offset	Parameter name	Parameter value
Write	21	ca_url	string. Each url is 0-256 characters. The three parameters can not be all empty.
		client_cert_url	
		client_key_url	

Setup:

```
{
  "msg_id": 1021,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "ca_url": "http://47.104.172.255:8080/ca.pem",
    "client_cert_url": "http://47.104.172.255:8080/client_cert.pem",
    "client_key_url": "http://47.104.172.255:8080/client_key.key"
  }
}
```

Device reply:

```
{
  "msg_id": 1021,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.2.3 EAP certificate download result (022)

To gateway will actively notify the result after it downloads the EAP certificates.

Flag	Msg id offset	Parameter name	Parameter value
Notify	22	result_code	number. -1: under process 0: failed 1: success

Device notify:

```
{
  "msg_id": 3022,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "result_code": 1
  }
}
```

3.3.2.4 IP setting (023)

To inquire and change the ip settings.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	23	dhcp_en	number. 0-1 0: disable 1: enable
		ip	string. 7-15 characters
		netmask	string. 7-15 characters
		gw	string. 7-15 characters
		dns	string. 7-15 characters

Inquire:

```
{
  "msg_id": 2023,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2023,
  "device_info": {
    "mac": "4c11ae8be624"
  },
}
```

```

    "data": {
      "dhcp_en": 1,
      "ip": "0.0.0.0",
      "netmask": "0.0.0.0",
      "gw": "0.0.0.0",
      "dns": "0.0.0.0"
    }
  }
}

```

Setup:

```

{
  "msg_id": 1023,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "dhcp_en": 1,
    "ip": "192.169.0.10",
    "netmask": "255.255.255.0",
    "gw": "192.168.0.1",
    "dns": "8.8.8.8"
  }
}

```

Device reply:

```

{
  "msg_id": 1023,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}

```

3.3.3 MQTT settings

To inquire and change the MQTT settings for the gateway. After sending these commands, and send the reboot command (msg id: 1000), the gateway will reboot and reconnect using the new MQTT settings.

3.3.3.1 MQTT parameters (030)

Flag	Msg id offset	Parameter name	Parameter value
Write/read	30	security_type	number. 0-3

			0: no encryption 1: CA signed server certificate 2: CA certificate 3: Self signed certificates
		host	string. 1-64 characters
		port	number. 1-65535
		client_id	string. 1-64 characters
		username	string. 0-256 characters
		passwd	string. 0-256 characters
		sub_topic	string. 1-128 characters
		pub_topic	string. 1-128 characters
		qos	number. 0-2
		clean_session	number. 0-1
		keepalive	number. 10-120, second
		lwt_en	number. 0-1
		lwt_qos	number. 0-2
		lwt_retain	number. 0-1
		lwt_topic	string. 1-128 characters
		lwt_payload	string. 1-128 characters

Inquire:

```
{
  "msg_id": 2030,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2030,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "security_type": 0,
    "host": "47.11.11.11",
    "port": 1883,
    "client_id": "4c11ae8be624",
    "username": ""
  }
}
```

```

        "passwd": "",
        "sub_topic": "/MK110/4c11ae8be624/receive",
        "pub_topic": "/MK110/4c11ae8be624/send",
        "qos": 0,
        "clean_session": 1,
        "keepalive": 60,
        "lwt_en": 0,
        "lwt_qos": 1,
        "lwt_retain": 0,
        "lwt_topic": "/MK110/4c11ae8be624/send",
        "lwt_payload": "offline"
    }
}

```

Setup:

```

{
    "msg_id": 1030,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "security_type": 0,
        "host": "47.11.11.11",
        "port": 1883,
        "client_id": "4c11ae8be624",
        "username": "",
        "passwd": "",
        "sub_topic": "/MK110/4c11ae8be624/receive",
        "pub_topic": "/MK110/4c11ae8be624/send",
        "qos": 0,
        "clean_session": 1,
        "keepalive": 60,
        "lwt_en": 0,
        "lwt_qos": 1,
        "lwt_retain": 0,
        "lwt_topic": "/MK110/4c11ae8be624/send",
        "lwt_payload": "offline"
    }
}

```

Device reply

```

{
    "msg_id": 1030,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}

```

```

    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.3.2 MQTT certificates (031)

To change the EAP certificates, valid when the WIFI security is enterprise.

Flag	Msg id offset	Parameter name	Parameter value
Write	31	ca_url	string.
		client_cert_url	Each url is 0-256 characters.
		client_key_url	The three parameters can not be all empty.

Setup:

```

{
  "msg_id": 1031,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "ca_url": "http://47.104.172.255:8080/ca.pem",
    "client_cert_url": "http://47.104.172.255:8080/client_cert.pem",
    "client_key_url": "http://47.104.172.255:8080/client_key.key"
  }
}

```

Device reply

```

{
  "msg_id": 1031,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}

```

3.3.3.3 MQTT certificate download result (032)

The gateway will actively notify the result after it downloads the MQT certificate.

Flag	Msg id offset	Parameter name	Parameter value
Notify	32	result_code	number. -1: under process 0: failed

			1: success
--	--	--	------------

Device notify:

```
{
  "msg_id": 3032,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "result_code": 1
  }
}
```

3.3.4 Bluetooth scan and advertising parameters

3.3.4.1 Scan switch (040)

To inquire and setup BLE scanning switch.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	40	scan_switch	number. 0-1 0: off 1: on

Inquire:

```
{
  "msg_id": 2040,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2040,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "scan_switch": 0
  }
}
```

Setup:

```
{
```

```

    "msg_id": 1040,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "scan_switch": 0
    }
}

```

Device reply:

```

{
    "msg_id": 1040,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.2 Filter relation (041)

To inquire and setup the relation of the MAC filter (043), adv name filter (044) and raw data filter (045). If the filters have been setup, but the relation does not include the filter, the filter will not take effect.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	41	relation	number. 0-7 0: none 1: mac 2: adv name 3: raw data 4: adv name& raw data 5: mac& adv name& raw data 6: adv name raw data 7: ADV name& mac

Inquire:

```

{
    "msg_id": 2041,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}

```

Device reply:

```

{

```

```

    "msg_id": 2041,
    "device_info": {
      "mac": "4c11ae8be624"
    },
    "data": {
      "relation": 4
    }
  }
}

```

Setup:

```

{
  "msg_id": 1041,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "relation": 4
  }
}

```

Device reply:

```

{
  "msg_id": 1041,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}

```

3.3.4.3 RSSI filter (042)

To inquire and setup the BLE RSSI to filter Beacon data.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	42	rsi	number. -127~0, dBm

Inquire:

```

{
  "msg_id": 2042,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}

```

Device reply:

```
{
  "msg_id": 2042,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "rssi": -127
  }
}
```

Setup:

```
{
  "msg_id": 1042,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "rssi": -127
  }
}
```

Device reply:

```
{
  "msg_id": 1042,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.4 Mac filter (043)

To inquire and setup the BLE MAC address filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	43	precise	number. 0-1 0: fuzzy filter 1: precision filter
		reverse	number. 0-1 0: positive filter 1: reverse filter
		mac	array. 0-10 group mac address, 0 means no filter. Each mac address is 2-12 characters (1-6 bytes).

Inquire:

```
{
  "msg_id": 2043,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2043,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "precise": 0,
    "reverse": 0,
    "mac": [ ]
  }
}
```

Setup:

```
{
  "msg_id": 1043,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "precise": 0,
    "reverse": 0,
    "mac": [
      "112233",
      "445566"
    ]
  }
}
```

Device reply:

```
{
  "msg_id": 1043,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

```
}
```

3.3.4.5 ADV name filter (044)

To inquire and setup the BLE advertising name filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	44	precise	number. 0-1 0: fuzzy filter 1: precision filter
		reverse	number. 0-1 0: positive filter 1: reverse filter
		name	array, 0-10 groups adv name, 0 means no filter. Each adv name is 1-20 characters.

Inquire:

```
{
  "msg_id": 2044,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2044,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "precise": 0,
    "reverse": 0,
    "name": [ ]
  }
}
```

Setup:

```
{
  "msg_id": 1044,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
```

```

        "precise": 0,
        "reverse": 0,
        "name": [
            "MK110",
            "MK107"
        ]
    }
}

```

Device reply:

```

{
    "msg_id": 1044,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.6 Raw data filter switch (045)

To inquire the filter switch of BLE raw data types.

Flag	Msg id offset	Parameter name	Parameter value
Read	45	ibeacon	number. 0: off, 1: on
		eddystone_uid	number. 0: off, 1: on
		eddystone_url	number. 0: off, 1: on
		eddystone_tlm	number. 0: off, 1: on
		bxp_devinfo	number. 0: off, 1: on
		bxp_acc	number. 0: off, 1: on
		bxp_th	number. 0: off, 1: on
		bxp_button	number. 0: off, 1: on
		bxp_tag	number. 0: off, 1: on
		pir	number. 0: off, 1: on
		other	number.

			0: off, 1: on
		mk_tof	number. 0: off, 1: on

Inquire:

```
{
  "msg_id": 2045,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2045,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "ibeacon": 1,
    "eddystone_uid": 1,
    "eddystone_url": 1,
    "eddystone_tlm": 1,
    "bxp_devinfo": 1,
    "bxp_acc": 1,
    "bxp_th": 1,
    "bxp_button": 1,
    "bxp_tag": 1,
    "pir": 1,
    "other": 1,
    "mk_tof": 1,
  }
}
```

3.3.4.7 iBeacon filter (046)

To inquire and set up the filter rule for iBeacon frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	46	switch_value	number. 0: off 1: on
		min_major	number. 0-65535
		max_major	number. 0-65535 max major >= min major

		min_minor	number. 0-65535
		max_minor	number. 0-65535 max minor >= min minor
		uuid	string. 0-32 characters (0-16 bytes). 0 means do not filter by uuid

Inquire:

```
{
  "msg_id": 2046,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2046,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "min_major": 0,
    "max_major": 0,
    "min_minor": 0,
    "max_minor": 0,
    "uuid": ""
  }
}
```

Setup:

```
{
  "msg_id": 1046,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "min_major": 0,
    "max_major": 0,
    "min_minor": 0,
    "max_minor": 0,
    "uuid": ""
  }
}
```

```
}
```

Device reply:

```
{
  "msg_id": 1046,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.8 Eddystone-UID filter (047)

To inquire and set up the filter rule for Eddystone-UID frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	47	switch_value	number. 0: off, 1: on
		namespace	string. 0-20 characters (0-10 bytes) 0 means do not filter by namespace
		instance	string. 0-12 characters (0-6 bytes) 0 means do not filter by instance

Inquire:

```
{
  "msg_id": 2047,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2047,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "namespace": "",
    "instance": ""
  }
}
```

```
}
```

Setup:

```
{
  "msg_id": 1047,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "namespace": "",
    "instance": ""
  }
}
```

Device reply:

```
{
  "msg_id": 1047,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.9 Eddystone-URL filter (048)

To inquire and set up the filter rule for Eddystone-URL frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	48	switch_value	number. 0: off, 1: on
		url	string. 0-37 characters 0 means do not filter by url

Inquire:

```
{
  "msg_id": 2048,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2048,
  "device_info": {
```

```

        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "url": ""
    }
}

```

Setup:

```

{
    "msg_id": 1048,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "url": "https://mokosmart.com"
    }
}

```

Device reply:

```

{
    "msg_id": 1048,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.10 Eddystone-TLM filter (049)

To inquire and set up the filter rule for Eddystone-TLM frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	49	switch_value	number. 0: off 1: on
		tlm_version	number. 0-2 0: version 0 1: version 1 2: all version

Inquire:

```

{
    "msg_id": 2049,

```

```

    "device_info": {
        "mac": "4c11ae8be624"
    }
}

```

Device reply:

```

{
    "msg_id": 2049,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "tlm_version": 0
    }
}

```

Setup:

```

{
    "msg_id": 1049,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "tlm_version": 0
    }
}

```

Device reply:

```

{
    "msg_id": 1049,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.11 BXP- device info filter (050)

To inquire and set up the filter rule for MOKO BXP-device info frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	50	switch_value	number. 0: off, 1: on

Inquire:

```
{
  "msg_id": 2050,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2050,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1
  }
}
```

Setup:

```
{
  "msg_id": 1050,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1
  }
}
```

Device reply:

```
{
  "msg_id": 1050,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.12 BXP-ACC filter (051)

To inquire and set up the filter rule for MOKO BXP-ACC frame.

Flag	Msg id offset	Parameter name	Parameter value
------	---------------	----------------	-----------------

Write/read	51	switch_value	number. 0: off, 1: on
------------	----	--------------	--------------------------

Inquire:

```
{
  "msg_id": 2051,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2051,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1
  }
}
```

Setup:

```
{
  "msg_id": 1051,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1
  }
}
```

Device reply:

```
{
  "msg_id": 1051,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.13 BXP-TH filter (052)

To inquire and set up the filter rule for MOKO BXP-TH frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	52	switch_value	number. 0: off, 1: on

Inquire:

```
{
  "msg_id": 2052,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2052,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1
  }
}
```

Setup:

```
{
  "msg_id": 1052,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1
  }
}
```

Device reply:

```
{
  "msg_id": 1052,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.14 BXP-Button filter (053)

To inquire and set up the filter rule for MOKO BXP-Button frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	53	switch_value	number. 0: off, 1: on
		single_press	number. 0: off, 1: on
		double_press	number. 0: off, 1: on
		long_press	number. 0: off, 1: on
		abnormal_inactivity	number. 0: off, 1: on

Inquire:

```
{
  "msg_id": 2053,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2053,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "single_press": 1,
    "double_press": 1,
    "long_press": 1,
    "abnormal_inactivity": 1
  }
}
```

Setup:

```
{
  "msg_id": 1053,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
```

```

        "switch_value": 1,
        "single_press": 1,
        "double_press": 1,
        "long_press": 1,
        "abnormal_inactivity": 1
    }
}

```

Device reply:

```

{
    "msg_id": 1053,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.15 BXP-Tag filter (054)

To inquire and set up the filter rule for MOKO BXP-Tag frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	54	switch_value	number. 0: off, 1: on
		precise	number. 0-1 0: fuzzy filter 1: precision filter
		reverse	number. 0-1 0: positive filter 1: reverse filter
		tagid	array. 0-10 group tag id. Each tag id is 2-12 characters (1-6 bytes).

Inquire:

```

{
    "msg_id": 2054,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}

```

Device reply:

```

{
    "msg_id": 2054,
    "device_info": {

```

```

        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "precise": 0,
        "reverse": 0,
        "tagid": [ ]
    }
}

```

Setup:

```

{
    "msg_id": 1054,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "precise": 0,
        "reverse": 0,
        "tagid": [
            "000",
            "111"
        ]
    }
}

```

Device reply:

```

{
    "msg_id": 1054,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.16 PIR presence filter (055)

To inquire and set up the filter rule for MOKO PIR frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	55	switch_value	number. 0: off, 1: on
		delay_response_status	number. 0-3 0: low delay

			1: medium delay 2: high delay 3: all types
		door_status	number. 0-2 0: close 1: open 2: all types
		sensor_sensitivity	number. 0-3 0: low sensitivity 1: medium sensitivity 2: high sensitivity 3: all types
		sensor_detection_status	number. 0-2 0: no motion detected 1: motion detected 2: all status
		min_major	number. 0-65535
		max_major	number. 0-65535 max major >= min major
		min_minor	number. 0-65535
		max_minor	number. 0-65535 max minor >= min minor

Inquire:

```
{
  "msg_id": 2055,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2055,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "delay_response_status": 3,
    "door_status": 2,
    "sensor_sensitivity": 3,
    "sensor_detection_status": 2,
    "min_major": 0,
```

```

        "max_major": 0,
        "min_minor": 0,
        "max_minor": 0
    }
}

```

Setup:

```

{
    "msg_id": 1055,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "delay_response_status": 3,
        "door_status": 2,
        "sensor_sensitivity": 3,
        "sensor_detection_status": 2,
        "min_major": 0,
        "max_major": 0,
        "min_minor": 0,
        "max_minor": 0
    }
}

```

Device reply:

```

{
    "msg_id": 1055,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.17 MK TOF filter (062)

To inquire and set up the filter rule for MOKO TOF frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	62	switch_value	number. 0: off, 1: on
		mfg_code	array. 0-10 group codes. Each code is 2 bytes.

Inquire:

```
{
  "msg_id": 2062,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2062,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "mfg_code": [ ]
  }
}
```

Setup:

```
{
  "msg_id": 1062,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "mfg_code": [
      "0059",
      "0078"
    ]
  }
}
```

Device reply:

```
{
  "msg_id": 1054,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.18 Other type filter (056)

To inquire and set up the filter rule for other type frame.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	56	switch_value	number. 0: off, 1: on
		relation	number. 0-5 0: A 1: A&B 2: A B 3: A&B&C 4: (A&B) C 5: A B C
		rule	array. Can set at most 3 groups filter rules.
		type	Data of the array. String, 2 characters (1 byte). BLE Data type, 00 means do not filter by data type.
		start	Data of the array. number: 0-29. The start byte under the data type.
		end	Data of the array. number: 0-29. The end byte under the data type, should be greater than the start byte.
		raw_data	Data of the array, string. The BLE raw data from the start to end byte under the data type.

Inquire:

```
{
  "msg_id": 2056,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2056,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

```

    },
    "data": {
        "switch_value": 1,
        "relation": 5,
        "rule": [ ]
    }
}

```

Setup:

```

{
    "msg_id": 1056,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "switch_value": 1,
        "relation": 0,
        "rule": [
            {
                "type": "00",
                "start": 1,
                "end": 2,
                "raw_data": "1234"
            }
        ]
    }
}

```

Device reply:

```

{
    "msg_id": 1056,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.19 Duplicate data filter (057)

To inquire and set up the rule to filter out duplicate BLE packets.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	57	rule	number. 0-3 0: no filter 1: mac

			2: mac+ data type 3: mac+ raw data
		timeout	number. 1-86400, second default: 10

Inquire:

```
{
  "msg_id": 2057,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2057,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "rule": 1,
    "timeout": 10
  }
}
```

Setup:

```
{
  "msg_id": 1057,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "rule": 1,
    "timeout": 10
  }
}
```

Device reply:

```
{
  "msg_id": 1057,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

```
}
```

3.3.4.20 Upload BLE scanning packet (059)

To inquire and setup the data content of BLE scanning packet.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	59	timestamp	number. 0-1 0: not upload, 1: upload
		adv_data	number. 0-1 0: not upload adverting packet 1: upload adverting packet
		rsp_data	number. 0-1 0: not upload response packet 1: upload response packet

Inquire:

```
{
  "msg_id": 2059,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2059,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "timestamp": 1,
    "adv_data": 1,
    "rsp_data": 1
  }
}
```

Setup:

```
{
  "msg_id": 1059,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "timestamp": 1,
    "adv_data": 1,

```

```

        "rsp_data": 1
    }
}

```

Device reply:

```

{
    "msg_id": 1059,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.4.21 PHY filter (060)

To inquire and setup the PHY filter.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	60	phy_filter	number. 0-3 0: 1M PHY (V4.2) 1: 1M PHY (V5.0) 2: 1M PHY (V4.2) & 1M PHY (V5.0) 3: Coded PHY (V5.0)

Inquire:

```

{
    "msg_id": 2060,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}

```

Device reply:

```

{
    "msg_id": 2060,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "phy_filter": 1,
    }
}

```

Setup:

```
{
  "msg_id": 1060,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "phy_filter": 1,
  }
}
```

Device reply:

```
{
  "msg_id": 1060,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.4.22 Advertise iBeacon (061)

To inquire and setup the iBeacon advertising parameters.

Flag	Msg id offset	Parameter name	Parameter value
Write/read	61	switch_value	number. 0-1 0: off, 1: on
		major	number. 0-65535
		minor	number. 0-65535
		uuid	string. 32 characters (16 bytes)
		adv_interval	number. 1-100, unit: 100ms
		rss_i_1m	number. -100~0, unit: dbm
		tx_power	number. 0-15 0: -24 dBm 1: -21 dBm 2: -18 dBm 3: -15 dBm 4: -12 dBm 5: -9 dBm 6: -6 dBm 7: -3 dBm 8: 0 dBm 9: 3 dBm 10: 6 dBm

			11: 9 dBm
			12: 12 dBm
			13: 15 dBm
			14: 18 dBm
			15: 21 dBm

Inquire:

```
{
  "msg_id": 2061,
  "device_info": {
    "mac": "4c11ae8be624"
  }
}
```

Device reply:

```
{
  "msg_id": 2061,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "major": 100,
    "minor": 99,
    "uuid": "00112233445566778899001122334455",
    "rssi_1m": -59,
    "adv_interval": 1,
    "tx_power": 9
  }
}
```

Setup:

```
{
  "msg_id": 1061,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "switch_value": 1,
    "major": 100,
    "minor": 99,
    "uuid": "00112233445566778899001122334455",
    "rssi_1m": -59,
    "adv_interval": 1,
    "tx_power": 9
  }
}
```

```
}
}
```

Device reply:

```
{
  "msg_id": 1061,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.5 Notify scanning BLE packet (3070)

When the gateway scans the Beacon data, it will group the data to packet and notify to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	70	timestamp	number. Unix timestamp, millisecond level.
		timezone	Step by half timezone.
		type_code	number. 0-10 0: ibeacon 1: eddystone-uid 2: eddystone-url 3: eddystone-tlm 4: bxp-devinfo 5: bxp-acc 6: bxp-th 7: bxp-button 8: bxp-tag 9: pir 10: other 11: MK TOF
		type	The description for type code
		rsi	number.
		connectable	number. 0-1 0: not connectable

			1: connectable
		mac	mac address of Beacon
		Beacon adv packet	See the following chapters

3.3.6.1 iBeacon packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686738414798,
    "timezone" : -24,
    "type_code" : 0,
    "type" : "ibeacon",
    "rssi" : -93,
    "connectable" : 0,
    "mac" : "665544332202",
    "uuid" : "e2c56db5dffb48d2b060d0f5a71096e0",
    "major" : 1,
    "minor" : 2,
    "rssi_1m" : -56
  } ]
}
```

3.3.6.2 Eddystone-UID packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741768642,
    "timezone" : -24,
    "type_code" : 1,
    "type" : "eddystone-uid",
    "rssi" : -81,
    "connectable" : 1,
    "mac" : "e942f7285a79",
    "namespace" : "adb143488994071c8d43",
    "instance" : "a00000000001",
    "rssi_0m" : 0
  } ]
}
```

3.3.6.3 Eddystone-URL packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741869955,
    "timezone" : -24,
    "type_code" : 2,
    "type" : "eddystone-url",
    "rssi" : -93,
    "connectable" : 0,
    "mac" : "ef5aaac187c0",
    "url" : "https://www.mokosmart.com",
    "rssi_0m" : 0
  } ]
}
```

3.3.6.4 Eddystone-TLM packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741926655,
    "timezone" : -24,
    "type_code" : 3,
    "type" : "eddystone-tlm",
    "rssi" : -75,
    "connectable" : 1,
    "mac" : "e0798d04a768",
    "tlm_version" : 0,
    "batt_vol" : 3023,
    "temperature" : 27.3,
    "adv_count" : 577,
    "runtime" : "17d9h49m34.2S"
  } ]
}
```

3.3.6.5 BXP-device info packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686741961892,
    "timezone" : -24,
    "type_code" : 4,
    "type" : "bxp-devinfo",
    "rssi" : -75,
    "connectable" : 1,
    "mac" : "f0c9b61e2854",
    "ranging_data" : -65,
    "adv_interval" : 1000,
    "batt_vol" : 2914,
    "passwd_verification" : 1,
    "have_light_sensor" : 0,
    "light_sensor_status" : 0,
    "firmware_ver" : "V3.3.7"
  } ]
}
```

3.3.6.6 BXP-ACC packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742061404,
    "timezone" : -24,
    "type_code" : 5,
    "type" : "bxp-acc",
    "rssi" : -82,
    "connectable" : 1,
    "mac" : "e23e4e4ae6b3",
    "ranging_data" : 2,
    "adv_interval" : 1000,
    "sampling_rate" : 10,
    "full_scale" : "2g",
    "motion_threshold" : 100,
    "x_axis_data" : 64,
    "y_axis_data" : -5056,
    "z_axis_data" : 15488,
    "batt_vol" : 3024
  } ]
}
```

3.3.6.7 BXP-TH packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742096971,
    "timezone" : -24,
    "type_code" : 6,
    "type" : "bxp-th",
    "rssi" : -84,
    "connectable" : 1,
    "mac" : "c3d7c95e2813",
    "txpower" : 0,
    "ranging_data" : 0,
    "adv_interval" : 500,
    "temperature" : 26.8,
    "humidity" : 56.4,
    "batt_vol" : 3447
  } ]
}
```

3.3.6.8 BXP-Button packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742166549,
    "timezone" : -24,
    "type_code" : 7,
    "type" : "bxp-button",
    "rssi" : -46,
    "connectable" : 1,
    "mac" : "e0b7f674b421",
    "frame_type" : 0,
    "passwd_verification" : 1,
    "alarm_status" : 0,
    "trigger_count" : 0,
    "device_id" : "000001",
    "adv_name" : "buttonliu",
    "full_scale" : "2g",
    "motion_threshold" : 16,
    "x_axis_data" : 0,
    "y_axis_data" : 0,
    "z_axis_data" : 0,
    "temperature" : 10.0,
    "ranging_data" : 0,
    "batt_vol" : 2808,
    "txpower" : 0
  } ]
}
```

3.3.6.9 BXP-Tag packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742201722,
    "timezone" : -24,
    "type_code" : 8,
    "type" : "bxp-tag",
    "rssi" : -93,
    "connectable" : 1,
    "mac" : "e0798d04b101",
    "hall_sensor_status" : 1,
    "have_accelerometer_sensor" : 1,
    "accelerometer_check_move" : 0,
    "hall_trigger_count" : 0,
    "motion_trigger_count" : 0,
    "x_axis_data" : -480,
    "y_axis_data" : -736,
    "z_axis_data" : 452,
    "batt_vol" : 3196,
    "tagid" : "000001"
  } ]
}
```

3.3.6.10 PIR packet

```
{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742244202,
    "timezone" : -24,
    "type_code" : 9,
    "type" : "pir",
    "rssi" : -88,
    "connectable" : 1,
    "mac" : "e8e77efac05b",
    "delay_response_status" : 0,
    "door_status" : 1,
    "sensor_sensitivity" : 0,
    "sensor_detection_status" : 0,
    "batt_vol" : 3504,
    "major" : 0,
    "minor" : 0,
    "rssi_1m" : -80
  } ]
}
```

3.3.6.11 MK TOF packet

```

{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "58bf253fb780"
  },
  "data" : [ {
    "timestamp" : 1747206222791,
    "timezone" : 0,
    "type_code" : 11,
    "type" : "mk-tof",
    "rssi" : -98,
    "connectable" : 1,
    "mac" : "fd1125997b0a",
    "mfg_code" : "0059",
    "beacon_code" : "6c07",
    "batt_vol" : 2997,
    "ranging_distance" : 19,
    "user_data" : "ffff",
    "sub_type" : 1
  } ]
}

```

3.3.6.12 Other type packet

```

{
  "msg_id" : 3070,
  "device_info" : {
    "mac" : "083a8d8bc9a0"
  },
  "data" : [ {
    "timestamp" : 1686742299971,
    "timezone" : -24,
    "adv_data" : "1eff060001092002945141029b8e0a0291313ea1298a0d09445c87dd4dac36",
    "rsp_data" : "",
    "type_code" : 10,
    "type" : "other",
    "rssi" : -40,
    "connectable" : 0,
    "mac" : "37196dc81666"
  } ]
}

```

3.3.6 Connect with MOKO Button

The gateway can connect to the MOKO Button beacon and send commands to manage the Button devices.

3.3.5.1 Connect MOKO Button device (100)

To make the gateway connect to the MK Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	100	mac	string. 12 characters (6 bytes of B1 mac address)
		passwd	string. 0-16 characters (0-8 bytes of password).

Setup:

```
{
  "msg_id": 1100,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "D9A5FA6F3882",
    "passwd": "Moko4321"
  }
}
```

Device reply:

```
{
  "msg_id": 1100,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.5.2 Notify connection result (101)

After the gateway connects the Button device, the gateway will actively notify the connection result. If connect successfully, the gateway will inquire the Button information and notify the information together.

Flag	Msg id offset	Parameter name	Parameter value
Write	101	result_code	number. 0-8 0: connect success 1: Gateway scanning is off

			2: Gateway is connected to other BLE device 3: The button is already connected 4: The button is out of range 5: The button is not connectable 6: Connect timeout 7: Device type error 8: Password error
		result_msg	string. The description for the result code
		product_model	string. Valid when result code is 0
		company_name	string. Valid when result code is 0
		hardware_version	string. Valid when result code is 0
		software_version	string. Valid when result code is 0
		firmware_version	string. Valid when result code is 0
		battery_v	number. unit:mV Valid when result code is 0
		single_alarm_num	number. Valid when result code is 0
		double_alarm_num	number. Valid when result code is 0
		long_alarm_num	number. Valid when result code is 0
		alarm_status	number.

```

{
  "msg_id": 3101,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "d9a5fa6f3882",
    "result_code": 0,
    "result_msg": "Connect succeed",
    "product_model": "BUTTON",
    "company_name": "MOKO TECHNOLOGY LTD.",
    "hardware_version": "MKBNseries",

```

```

        "software_version": "BXP-B-D",
        "firmware_version": "V1.0.1",
        "battery_v": 3000,
        "single_alarm_num": 18,
        "double_alarm_num": 0,
        "long_alarm_num": 0,
        "alarm_status": 1
    }
}

```

3.3.6.3 Inquire device information (102)

To inquire the device information of the Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	102	mac	string. 12 characters (6 bytes of B1 mac address)

Setup:

```

{
    "msg_id": 1102,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}

```

Device reply:

```

{
    "msg_id": 1102,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.5.3 Notify Button device information (103)

After sending the enquiry Button device information command, the gateway will get reply from the Button and notify the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value
------	---------------	----------------	-----------------

Notify	103	mac	string. 12 characters (6 bytes of B1 mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported
		product_model	string. Valid when the result code is 0
		company_name	string. Valid when the result code is 0
		hardware_version	string. Valid when the result code is 0
		software_version	string. Valid when the result code is 0
		firmware_version	string. Valid when the result code is 0

```
{
  "msg_id": 3103,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "d9a5fa6f3882",
    "result_code": 0,
    "result_msg": "Connect succeed",
    "product_model": "BUTTON",
    "company_name": "MOKO TECHNOLOGY LTD.",
    "hardware_version": "MKBNseries",
    "software_version": "BXP-B-D",
    "firmware_version": "V1.0.1"
  }
}
```

3.3.5.4 Inquire Button status (104)

To inquire the alarm status and battery status of the Button device.

Flag	Msg id offset	Parameter name	Parameter value
Write	104	mac	string. 12 characters (6 bytes of B1 mac address)

Setup:

```
{
  "msg_id": 1104,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "D9A5FA6F3882"
  }
}
```

Device reply:

```
{
  "msg_id": 1104,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "result_code": 0,
  "result_msg": "success"
}
```

3.3.5.5 Notify Button status (105)

After sending the enquiry Button device status command, the gateway will get reply from the Button and notify the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	105	mac	string. 12 characters (6 bytes of B1 mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported
		battery_v	number. unit:mV Valid when the result code is 0
		single_alarm_num	number. Valid when the result code is 0
		double_alarm_num	number. Valid when the result code is 0
		long_alarm_num	number. Valid when the result code is 0
		alarm_status	number. Valid when the result code is 0

```
{
```

```

    "msg_id": 3105,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0,
        "battery_v": 3000,
        "single_alarm_num": 18,
        "double_alarm_num": 0,
        "long_alarm_num": 0,
        "alarm_status": 1
    }
}

```

3.3.5.6 Dismiss alarm (106)

When the Button alarm is triggered, send command to the gateway to dismiss the alarm status of the Button.

Flag	Msg id offset	Parameter name	Parameter value
Write	106	mac	string. 12 characters (6 bytes of B1 mac address)

Setup:

```

{
    "msg_id": 1106,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}

```

Device reply:

```

{
    "msg_id": 1106,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.5.7 Notify alarm dismiss result (107)

After sending the dismiss alarm command, the gateway will get reply from the Button and notify the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	107	mac	string. 12 characters (6 bytes of B1 mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported

```
{
  "msg_id": 3107,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "d9a5fa6f3882",
    "result_code": 0
  }
}
```

3.3.5.8 Notify BLE disconnection (108)

When the gateway disconnects from the Button device, it will notify the reason to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	108	mac	string. 12 characters (6 bytes of B1 mac address)
		reason	number. 0-1 0: disconnect due to abnormal 1: disconnect due to communicate timeout

```
{
  "msg_id": 3108,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
```

```

        "mac": "d9a5fa6f3882",
        "reason": 0
    }
}

```

3.3.5.9 Control Beacon LED (109)

To control the Beacon LED via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	109	mac	string. 12 characters (6 bytes of Button mac address)
		flash_time	number. 1-6000, unit: 0.1 second
		flash_interval	number. 0-10000, unit: 0.1 second

Setup:

```

{
    "msg_id": 1109,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "flash_time": 100
        "flash_interval": 10
    }
}

```

Device reply:

```

{
    "msg_id": 1109,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.5.10 LED controlling result(110)

After sending the LED controlling command, the gateway will get reply from the Button and notify the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value
------	---------------	----------------	-----------------

Notify	110	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported

```
{
  "msg_id": 1110,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "d9a5fa6f3882",
    "result_code": 0
  }
}
```

3.3.5.11 Control Beacon Buzzer (111)

To control the Beacon Buzzer via gateway.

Flag	Msg id offset	Parameter name	Parameter value
Write	111	mac	string. 12 characters (6 bytes of Button mac address)
		ring_time	number. 1-6000, unit: 0.1 second
		ring_interval	number. 0-10000, unit: 0.1 second

Setup

```
{
  "msg_id": 1111,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "d9a5fa6f3882",
    "ring_time": 100
    "ring_interval": 10
  }
}
```

Device reply:

```
{
```

```

    "msg_id": 1111,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.5.12 LED controlling result(112)

After sending the Buzzer controlling command, the gateway will get reply from the Button and notify the reply to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	112	mac	string. 12 characters (6 bytes of Button mac address)
		result_code	number. 0-2 0: success 1: the Button is not connected by gateway 2: command is not supported

```

{
    "msg_id": 1112,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "d9a5fa6f3882",
        "result_code": 0
    }
}

```

3.3.5.13 Disconnect BLE device (200)

To setup the gateway to disconnect from the Button device.

Flag	Msg id offset	Parameter name	Parameter value
write	200	mac	string. 12 characters (6 bytes of B1 mac address)

Setup:

```

{
    "msg_id": 1200,
    "device_info": {

```

```

        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882"
    }
}

```

Device reply:

```

{
    "msg_id": 1200,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.5.14 Gateway connect status (201)

To inquire the gateway and Beacon connection status.

Flag	Msg id offset	Parameter name	Parameter value
Read	201	ble_conn_list	array
		mac	The mac of BLE devices connected by gateway
		type	number. 0: other BLE device 1: MKButton

Inquire:

```

{
    "msg_id": 2201,
    "device_info": {
        "mac": "4c11ae8be624"
    }
}

```

Device reply:

```

{
    "msg_id": 2201,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "ble_conn_list": [

```

```

        {
            "mac": "d9a5fa6f3882",
            "type": 1
        }
    ]
}

```

3.3.5.15 DFU Beacon firmware (202)

To setup the gateway upgrade Beacon firmware. After sending this command, the gateway will download the firmware files from the certain HTTP server, and transfer the files to the Button for upgrading.

Flag	Msg id offset	Parameter name	Parameter value
Write	202	mac	string. 12 characters BLE device MAC address
		firmware_url	string. 1-256 characters
		init_data_url	string. 1-256 characters

Setup:

```

{
    "msg_id": 1202,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "data": {
        "mac": "D9A5FA6F3882",
        "firmware_url": "http://47.104.172.255:8080/updata_fold/BXP_B_APP_1.0.1.bin",
        "init_data_url": "http://47.104.172.255:8080/updata_fold/BXP_B_APP_1.0.1.dat"
    }
}

```

Device reply:

```

{
    "msg_id": 1202,
    "device_info": {
        "mac": "4c11ae8be624"
    },
    "result_code": 0,
    "result_msg": "success"
}

```

3.3.5.16 Notify DFU process (203)

When the beacon id under DFU, the gateway will inquire the DFU process and notify to cloud every second.

Flag	Msg id offset	Parameter name	Parameter value
Notify	203	mac	string. 12 characters BLE device MAC address
		percent	number.

```
{
  "msg_id": 3203,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "d9a5fa6f3882",
    "percent": 20
  }
}
```

3.3.5.17 Notify DFU result (204)

After the beacon DFU is completed, the gateway will notify the result to cloud.

Flag	Msg id offset	Parameter name	Parameter value
Notify	204	mac	string. 12 characters BLE device MAC address
		result_code	number. 0-5 0: success 1: BLE device is not connected 2: command is not supported 3: Failed to get the upgrade files 4: command timeout 5: fail

```
{
  "msg_id": 3204,
  "device_info": {
    "mac": "4c11ae8be624"
  },
  "data": {
    "mac": "d9a5fa6f3882",
    "result_code": 0
  }
}
```

4. Revision History

Revision	Description	Editor	Date
V1.0	Initial Version	Weiguifen	2025.5.14

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